My Data Literacy Project
(Replace this with your Project Title)

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Abstract

Put your abstract here. Abstracts typically start with a sentence motivating why the subject is interesting. Then mention the data, methodology or methods you are working with, and describe results.

1. Data and Methods

NOTE: This is also Data Cleaning For the analysis of performance evolution and performance density evolution we made some restrictions to the dataset to have comparable and sufficient data. We analysed for the U18, U20 and Adult age groups, representing the most relevant age groups with consistent data. U23 didn't had enough data, also for younger age groups. Also always used the top 35 Ranks for analysis to have enough data for all disciplines and age groups. A Group is defined as a combination of Gender x Age group x Discipline to ensure consistency with the original leaderboards.

There were changes in weights and heights in the period of our data. In the U20 category some throwing weights and the height of 110 m hurdles changed (see (**?**)). We observed changes in 2012 in the weights of U18 categories. The results of the median and rank analyses were annulled, as these results are solely attributable to the change. The results of the performance gap analysis were retained, as these are independent of absolute performance levels.

1.1. Performance evolution analysis

To analyze long-term structural changes, we compare the five-year periods 2001–2005 and 2021–2025. This timeframe smooths out annual fluctuations and, through a broader data base, ensures the statistical validity of the results. The analysis is conducted on two levels: 1. Recording performance trends and density for all groups to identify group differences. 2. Examining the development to identify the structure in which performance changes occur. This allows us to identify general trends and determine the structure of the changes.

To assess general changes in performance, the percentage difference between the medians is used, as the median represents the typical performance level and is more robust against outliers compared to the mean. Medians are determined for each group from the data pools of the respective five-year periods (2001–2005 vs. 2021–2025). The development is then calculated as a percentage difference, normalized to the baseline period. A positive value indicates an improvement in performance, while a negative value represents a decline. Statistical significance is tested using the Mann-Whitney U test, which allows for a comparison of the periods without assuming a normal distribution.

To measure the performance density, we define the performance gap (G_p) between the three best and the three worst (e.g., 33rd–35th place) in our data set for the two time periods. The averaging of the top 3 reflects the international standard of success (podium), while the bottom 3 provide a stable and comparable baseline. We used the performance gap to measure the breadth of performance at the top level and utilized all available information (e.g., rankings) from our dataset to measure polarization without assuming a normal distribution. Mathematically, the gap (G_p) is defined as follows:

$$G_p = \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=1}^{3} x_{y,r} \right) - \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=33}^{35} x_{y,r} \right) \tag{1}$$

where $x_{y,r}$ represents the individual performance of rank r in year y, and $|P|$, in our case $|P| = 5$, represents the number of years per period. The development of the performance gap was calculated as a percentage change between the start and end periods, normalized to the start period. A positive value indicates a decrease and a negative value indicates an increase in the performance gap.

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Statistical significance was determined using bootstrap resampling with B = 10,000 iterations. This is a non-parametric method and is well-suited for the small sample size (N = 12) in our case. By drawing with replacement from the performance pools, we generated a distribution of the differences in performance gaps to calculate the 95% confidence interval (CI). If the CI contains zero, the null hypothesis that the performance density does not change significantly cannot be rejected.

To analyze the development of the different rankings, we calculate the arithmetic mean for each rank across all groups for the start and end periods. For each rank, even when differentiating by age group and gender, the sample size is at least n = 80. According to the central limit theorem, we can assume a normal distribution and thus calculate the arithmetic mean. The development of each rank was determined by calculating the percentage change between the start and end periods, normalized to the start period. A positive value indicates an improvement in the average ranking, a negative value a decline.

2. Results

Figure 1 shows the change in performance spread and typical performance development from 2021-2025 compared to 2001-2005 in many groups. The analysis of the performance data encompasses 106 groups, where X groups were excluded for median calculations (see 1. Overall, a decrease in performance density is evident. The best performers are improving, while the weaker performers are declining. While median performance trends were nearly balanced with 38 groups improving and 37 declining, the performance spread shows a strong asymmetry: A significant expansion was observed in 49.1% (52 groups), while a reduction was only seen in 4.7% (6 groups). This widening of the performance gap was most pronounced among men, especially in the U20 (70.6%) age group. In contrast, the women's groups showed significantly greater average progress: 63.3% improved compared to only 36.5% of the men's groups. Furthermore, discipline-specific trends emerged: The running and sprinting disciplines showed the most significant progress, with an improvement-to-decline ratio of 3:1 and 2:1 respectively. In contrast, the technical disciplines showed a significant decline: the declines outweighed the improvements by a ratio of 6:1 in throwing and 2.6:1 in jumping.

Figure 2 illustrates a linear polarization of performance. While the top-ranked company improved by an average of 1.61%, the 35th-rank declined by 1.51%, with the turning point towards negative growth being reached at the 22nd-rank. This polarization is not apparent only among adult women; all ranks from third place onward improved more than the top two ranks. Overall, women in all age groups perform better than their male counterparts. The most significant declines were seen in the male age groups under 18 and under 20. In the under-18 age group, the decline begins as early as rank 6, while in the under-20 age group it starts as early as rank 9.

3. Discussion & Conclusion

Discussion on Performance Spread

The results indicate a systematic increase in performance dispersion, particularly among men and young athletes. A major reason for this could be the 11.9% decline in DLV memberships (2005–2024) (Lei), with a more pronounced decrease of 26.6% in the 27-to 60-year-old age group. This reflects a decline in the number of coaches, who are often found in this age group. Technical disciplines, in particular, are feeling the effects due to their higher reliance on individual athletes, which could also explain their declining performance density. In contrast, women's athletics shows more improvements, which is probably due to a "catch-up effect" as a result of historically delayed professionalization.

In running competitions, the overall improvement in average performance was significantly influenced by the introduction of carbon fiber shoes in 2016 (Willwacher et al.). Furthermore, the COVID-19 pandemic led to uneven development. Endurance athletes may benefited from focused, uninterrupted training blocks (**?**). In contrast, athletes in technical disciplines were hindered by the closure of training facilities.

The polarization and the linear trend (see Fig X) could also be explained by the overall professionalization of the sport... TODO

Limitations

One limitation is the focus on the top 35. We cannot make any statements about the general development in all of amateur sports. The analysis only considers advanced competitive sports. Whether the observed trend continues linearly would need to be investigated with further analyses.

Conclusion

The increase in the performance spread indicates a structural change in organized athletics. A future focus should be placed on strengthening the grassroots level. Following the principle of "the grassroots supporting the elite," this will allow for more top performances and could bring German athletics closer to the world's elite.

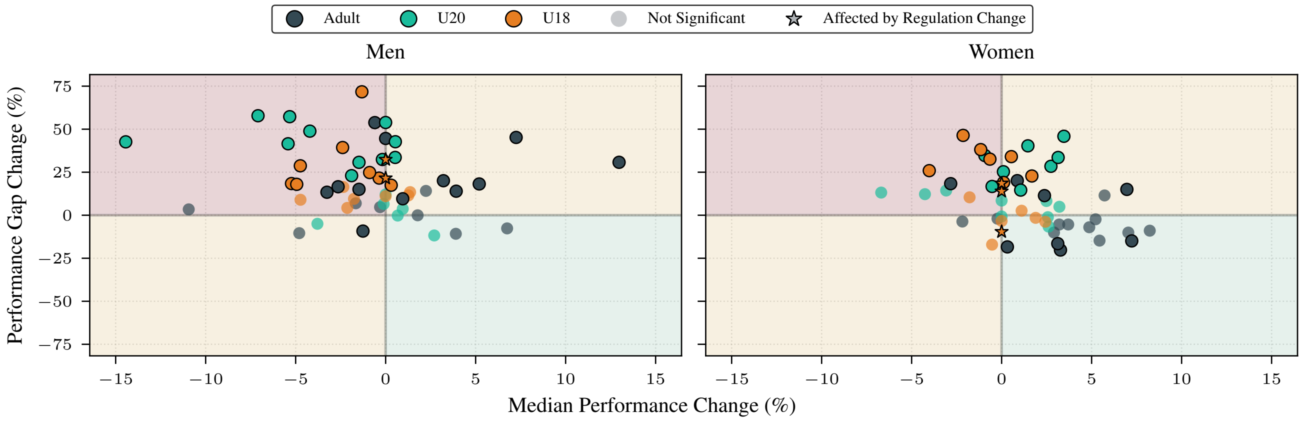

Figure 1. Relationship between median performance and change in performance gap (2001–2005 to 2021–2025). The scatter plot shows the percentage change in median performance (x-axis) and the change in performance gap (y-axis) for male (left) and female (right) groups, categorized by age group. The transparency of the markers indicates the statistical significance of the change in performance gap (via bootstrap method). Disciplines affected by regulatory changes are centered at 0 for the median, as only performance gap remains comparable across the time periods.

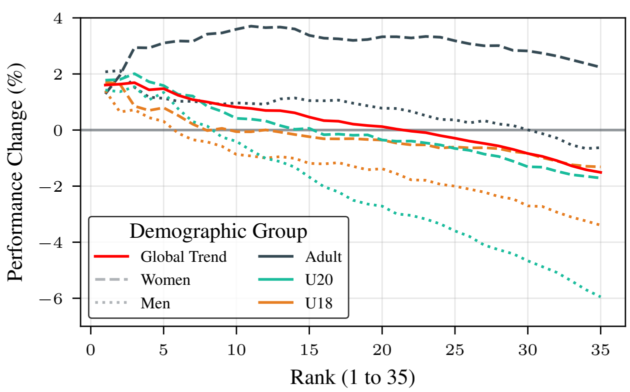

Figure 2. Performance development across ranks 1–35 (2001–05 to 2021–25). Shown is the percentage change in the average performance values of all groups overall, as well as broken down by age group and gender.

References

das leichtathletik-portal - downloads. URL https://www.leichtathletik.de/service/downloads.

Willwacher, S., Mai, P., Helwig, J., Hipper, M., Utku, B., and Robbin, J. *Does advanced footwear technology improve track and road racing performance? An explorative analysis based on the 100 best yearly performances in the world between 2010 and 2022*. doi: 10.51224/SRXIV.297.